Pre-Study Parameter Configuration

The selected NPR shader combines multiple rendering techniques in a single pass: an inverted hull geometry outline, Gaussian pre-filtered Sobel edge detection, XToon 2D ramp tonal shading, and rim light. Post-process effects including a Kuwahara painterly filter and a hierarchical depth edge layer can be added on top.

Although the shader includes a Gaussian pre-filter stage that smooths the luminance signal before Sobel edge detection, this stage was disabled across all pre-study configurations. Preliminary testing showed no perceptually meaningful visual difference between the pre-filter enabled and disabled at the parameter ranges used for this avatar, so a dedicated round for this parameter was omitted.

Each round varied one parameter group while all others were held constant. Participants ranked images twice per round: by stylization quality as a character, and by trustworthiness as an advisor.

Round 1 — Inner Line Threshold (L1–L5)

Label	`_InnerLineThreshold`
L1	0.01
L2	0.02
L3	0.03
L4	0.04
L5	0.05

Fixed: Gaussian blur off, normal edges off, inner line sample distance 0.2, light sensitivity 0.5, shadow strength 0.5, outline width 0.005, depth offset on, rim light on.

Round 2 — Outer Outline Width and Depth Offset (O1–O6)

Label	**Outline Width**	**Depth Offset**	**Depth Bias**
O1	0.005	Off	15
O2	0.005	On	15
O3	0.007	Off	15
O4	0.007	On	15
O5	0.003	Off	15
O6	0.003	On	15

Fixed: inner line threshold 0.05 (L5). Depth bias 15 throughout to make on/off difference clearly visible (effect is negligible below 5).

Round 3 — XToon Toon Shading (T1–T5)

Detail mode fixed to Depth. Curvature and Manual excluded (near-identical output at 1 m framing).

Label	Shadow Strength	Detail Bias
T1	1.0	0.5
T2	0.0	0.5
T3	0.5	0.5
T4	0.5	1.0
T5	0.5	0.0

Fixed: outline width 0.005, depth offset on (bias 15), inner line threshold 0.03.

Round 4 — Rim Light On/Off (R1–R2)

Rim power fixed at 4.0 (selected by preliminary testing as balanced without over-brightening the silhouette).

Label	Rim Light	Rim Power
R1	Off	4.0
R2	On	4.0

Round 5 — Kuwahara Kernel Size (K1–K4)

Kuwahara applied on top of V4 base (L5 configuration). K1 is the unfiltered V4 reference. Sector count excluded (no perceptible difference across its range). Tested values: 2, 5, 8. No visible difference was observed above kernel size 8.

Label	Kernel Size	Notes
K1	—	V4 baseline, no Kuwahara
K2	2	Minimal smoothing
K3	5	Mid-range
K4	8	Maximum visible smoothing

Round 6 — Kuwahara Hardness and Sharpness (A1–A4)

Applied on top of V4 base (L5). A1 is the V4 reference without Kuwahara. Hardness and sharpness are varied inversely.

Label	Hardness	Sharpness
A1	—	—
A2	1	18
A3	9	9
A4	18	1

Round 7 — Hierarchical Depth Edge Threshold (H1–H6)

Only depth edges tested. Colour edges already handled by V4 Sobel. Normal edges excluded (camera-angle-dependent, unsuitable for fixed-view ranking). Depth scale fixed at 2.

Label	Depth Threshold
H1	*(to be filled)*
H2	*(to be filled)*
H3	*(to be filled)*
H4	*(to be filled)*
H5	*(to be filled)*
H6	*(to be filled)*

Baseline V4 Configuration (L5, used as reference in Rounds 5–7)

Parameter	Value
Inner line threshold	0.05
Inner line sample distance	0.2
Gaussian blur	Off
Normal edges	Off
Fresnel edges	Off
Light sensitivity	0.5
Shadow strength	0.5
Detail mode	Depth
Detail bias	0.5
Outer outline width	0.005
Depth offset	On
Depth bias	15
Rim light	On
Rim power	4.0